

Table: QKS-v2 vs ViT4Mal [2] – IMG_DS on PYNQ-Z2 (3,444 test images)

Configuration	Acc (%)	F1	FPGA Model Inference Time	LUT	DSP	BRAM	vs ViT4Mal 0.81 s	vs ViT4Mal 1.48 s
Q16-QKSv2-S0 (dense)	93.64	96.2100	1.552 ms	13,322	147	15	522×	954×
-QKSv2-S10 (12.5% pr	93.79	96.3100	1.533 ms	13,344	147	15	528×	965×
QKSv2-S20 (18.75% p	94.11	96.5100	1.502 ms	13,349	148	15	539×	985×
QKSv2-S30 (31.25% p	93.84	96.3400	1.442 ms	13,347	148	15	562×	1026×
ViT4Mal 1e2h (fastest	810000.00	—	0 ms	16,545	69	92	8765.3×	16015.6×
ViT4Mal 1e4h	870000.00	—	0 ms	16,029	69	84	8759.6×	16005.2×
ViT4Mal 2e2h	1450000.00	—	0 ms	16,219	69	92	8629.9×	15768.2×
ViT4Mal 2e4h (best acc	1480000.00	—	0 ms	16,079	69	84	8627.1×	15763.1×